Exp-7 Creating a to-do list using Angular

src/

└── app/

├── app.ts

├── app.html

├── app.css

├── app.config.ts

├── todo.model.ts

├── todo.service.ts

├── todo/

│ ├── todo.component.ts

│ ├── todo.component.html

│ └── todo.component.css

Main.ts

```
import { bootstrapApplication } from '@angular/platform-browser';
import { App } from './app/app';
import { appConfig } from './app/app.config';
bootstrapApplication(App, appConfig)
  .catch(err => console.error(err));
```

App.config

```
import { ApplicationConfig } from '@angular/core';
export const appConfig: ApplicationConfig = {
  providers: []
};
```

App.ts

```
import { Component } from '@angular/core';
import { TodoComponent } from './todo/todo.component';

@Component({
  selector: 'app-root',
  standalone: true,
  imports: [TodoComponent],
  templateUrl: './app.html',
  styleUrls: ['./app.css']
})
export class App {
  title = 'To-Do List';
}
```

Todo.service.ts

```
import { Injectable } from '@angular/core';
import { Todo } from './todo.model';

@Injectable({
  providedIn: 'root'
})
export class TodoService {
  private todos: Todo[] = [
    {
      id: 1,
      title: 'Buy groceries',
      completed: false
    },
    {
      id: 2,
      title: 'Call the plumber',
```

```
    completed: false
  },
  {
    id: 3,
    title: 'Go for a walk',
    completed: false
  }
];
private nextId: number = 4;
constructor() {}
getTodos(): Todo[] {
  return this.todos;
}
addTodo(title: string): void {
  const newTodo: Todo = {
    id: this.nextId++,
    title: title,
    completed: false
  };
  this.todos.push(newTodo);
}
toggleTodoCompleted(id: number): void {
  const todo = this.todos.find(t => t.id === id);
  if (todo) {
    todo.completed = !todo.completed;
  }
}
deleteTodo(id: number): void {
  this.todos = this.todos.filter(t => t.id !== id);
}
}
```

Todo.model.ts

```
export interface Todo {
  id: number;
  title: string;
  completed: boolean;
}
```

Todo.component.ts

```
import { Component, OnInit } from '@angular/core';
import { CommonModule } from '@angular/common';
import { FormsModule } from '@angular/forms';
import { TodoService } from '../todo.service';
import { Todo } from '../todo.model';

@Component({
  selector: 'app-todo',
  standalone: true,
  imports: [CommonModule, FormsModule],
  templateUrl: './todo.component.html',
  styleUrls: ['./todo.component.css']
})
export class TodoComponent implements OnInit {
  todos: Todo[] = [];
  newTodoTitle: string = '';
  constructor(private todoService: TodoService) {}
  ngOnInit(): void {
    this.todos = this.todoService.getTodos();
  }

  addTodo(): void {
```

```
    if (this.newTodoTitle.trim()) {
      this.todoService.addTodo(this.newTodoTitle.trim());
      this.newTodoTitle = '';
      this.todos = this.todoService.getTodos();
    }
  }
  toggleTodoCompleted(id: number): void {
    this.todoService.toggleTodoCompleted(id);
    this.todos = this.todoService.getTodos();
  }
  deleteTodo(id: number): void {
    this.todoService.deleteTodo(id);
    this.todos = this.todoService.getTodos();
  }
}
```

Todo.component.html

```
<div class="todo-container">
  <h1>To-Do List</h1>
  <div class="add-todo">
    <input type="text"
      [(ngModel)]="newTodoTitle"
      placeholder="Enter a task"
      (keyup.enter)="addTodo()"  />
    <button (click)="addTodo()">
      Add
    </button>
  </div>
  <ul>
    <li *ngFor="let todo of todos">
      <input type="checkbox"    [checked]="todo.completed"
(change)="toggleTodoCompleted(todo.id)"/>
      <span [class.completed]="todo.completed">  {{ todo.title }}
```

```
      </span>
      <button class="delete-btn" (click)="deleteTodo(todo.id)">
        Delete
      </button>
    </li>
  </ul>
</div>
```

output

